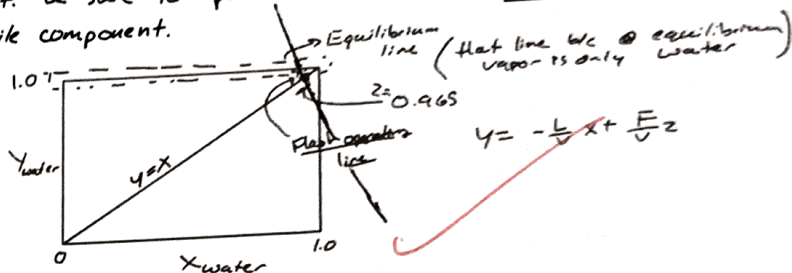
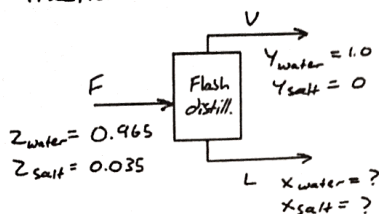


1) Problem 2.A4

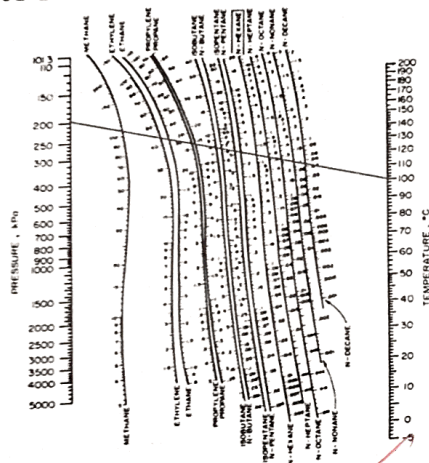
In the flash distillation of salt water, the salt is totally nonvolatile. Show a McCabe-Thiele diagram for feed water containing 3.5 wt% salt. Be sure to plot wt fraction of the more volatile component.



2) Problem 2.A8

Use the DePriester Chart

A) At 100°C & 200 kPa, what is k value of n-hexane?



$k = 1.2$

B) As pressure increases, does k value increase, decrease, or stay constant?

b) Decreases

[see trend on chart above]

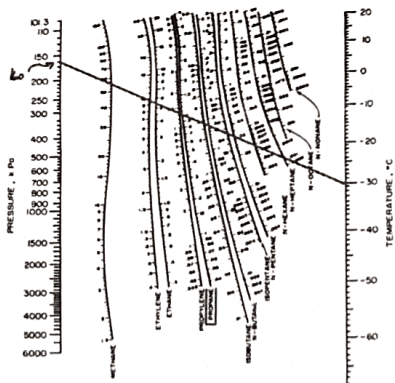
C) Within a homogeneous series such as light hydrocarbons, as the molecular weight increases, does k value (@ constant P & T) a) increase, b) decrease, or c) stay the same

b) Decreases

[compare hexane w/ decane on chart above]

D) At what pressure does pure propane boil @ temp. of -30°C?

where $k=1$ $\therefore$ draw line from $k=1$ & $T=-30^\circ\text{C}$ to find correct pressure.



$P = 160 \text{ kPa}$

3) Problem 2.A10

Use the vapor-liquid equilibrium data @ 1.0 atm for methanol-water (table 2-7) For the following:

A) If the methanol vapor mole fraction is 0.60, what is the methanol liquid mole fraction?

Methanol Liquid	Methanol Vapor	Temp., °C
0	0	100
2.0	13.4	96.4
4.0	23.0	93.5
6.0	30.4	91.2
8.0	36.5	89.3
10.0	41.8	87.7
15.0	51.7	84.4
20.0	57.9	81.7
30.0	66.5	78.0
40.0	72.9	75.3
50.0	77.9	73.1
60.0	82.5	71.2
70.0	87.0	69.3
80.0	91.5	67.6
90.0	95.8	66.0
95.0	97.9	65.0
100.0	100.0	64.5

Using table 2-7, create a temperature composition diagram for methanol-water [Excel]

$$X_{\text{methanol}} = 0.22$$

B) Is there an azeotrope in the methanol-water system?

No, can be tested by adding X_M & Y_M and seeing if any of the points are =1.

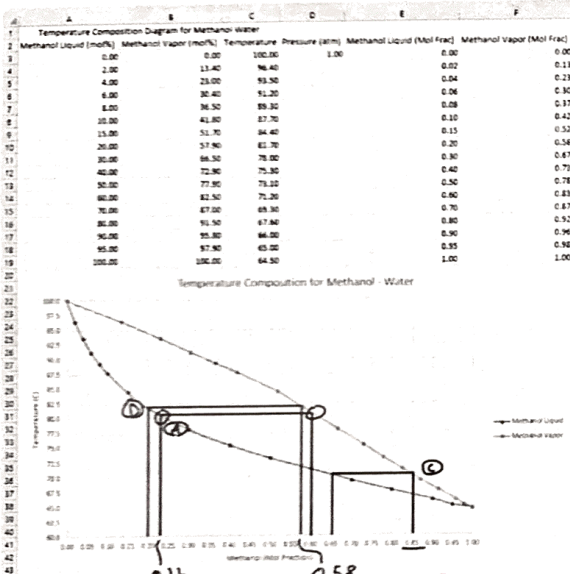
C) If water liquid mole fraction is 0.35, what is the water vapor mole fraction?

$$\text{If } X_W = 0.35 \rightarrow \text{then } X_M = 1 - X_W \therefore X_M = 0.65$$

Now use excel graph w/ $X_M = 0.65$ to find Y_M

$$Y_M = 0.85 \therefore Y_W = 1 - Y_M \Rightarrow Y_W = 0.15$$

Azeotrope Test
0.00
0.15
0.27
0.36
0.45
0.52
0.67
0.78
0.97
1.13
1.28
1.43
1.57
1.72
1.86
1.93
2.00



wrong graph!

D) What is the k value for methanol & of water at a methanol mol frac in the liquid of 0.20?

$X_M = 0.20$ using excel graph we can

$X_W = 0.80$ find Y_M then w/

$$Y_M = 0.58 \quad Y_W = 1 - Y_M \quad \& \quad k_i = \frac{y_i}{x_i}$$

$Y_W = 0.42$ to find k values.

$$k_M = \frac{0.58}{0.20} = 2.9, \quad k_W = \frac{0.42}{0.80} = 0.53$$

$$k_M = 2.9, \quad k_W = 0.53$$

E) What is the relative volatility α_{M-W} of a methanol mole fraction in the liquid of 0.20?

$X_M = 0.20 \therefore$ same k_M & k_W as part D. ($k_W = 0.53$)

$$\alpha_{M-W} = \frac{k_M}{k_W} = \frac{2.9}{0.53}$$

$$\alpha_{M-W} = 5.5$$

4) Problem 2.D1(a-d)

We are separating a mixture of methanol & water in a flash drum at 1.0 atm. See equilibrium data on table.

A) Feed is 60.0 mol% methanol & 40.0 mol% of feed is vaporized. What are the vapor & liquid mole fractions & flow rates? Feed is 100 kmol/h

$$\text{So } Z = 0.60, \quad F = 100 \frac{\text{kmol}}{\text{h}} \therefore \text{if 40\% vaporized then } V = 40 \frac{\text{kmol}}{\text{h}} \quad \& \quad L = 60 \frac{\text{kmol}}{\text{h}}$$

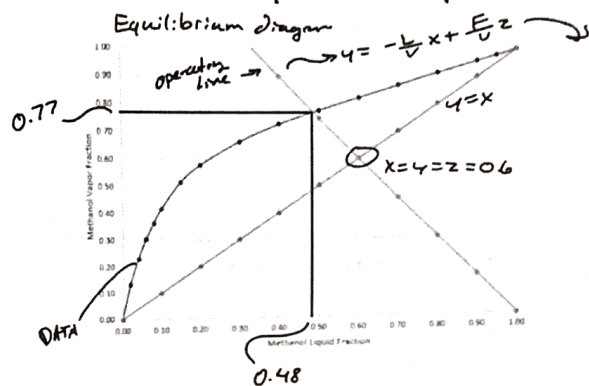
We can plot the equilibrium diagram from the given

step 1: data points & the $y=x$ line.

step 2: We know that $x=y=Z=0.60$ (b/c 60% mol methanol) so we know the operating line must intersect w/ this point.

step 3: Using $y = -\frac{L}{V}x + \frac{F}{V}Z$, we can find this line. we know L, V, Z , & F from above.

Step 1 Plot this line on the graph & mark the intersection of the data point curve to determine the mol fractions of methanol in the liquid & vapor.



$$y = -\frac{60}{40}x + \frac{100}{40}(0.6)$$

$$\left\{ y = -\frac{3}{2}x + \frac{3}{2} \right\}$$

$$y_M = 0.77 \text{ \& } x_M = 0.48$$

$$V = 40 \text{ kmol/h}$$

$$L = 60 \text{ kmol/h}$$

$$x_M = 0.48, x_W = 0.52$$

$$y_M = 0.77, y_W = 0.23$$

B) Repeat part A for a feed rate of $1600.0 \frac{\text{kmol}}{\text{h}}$

$$F = 1500.0 \frac{\text{kmol}}{\text{h}}$$

$$\begin{cases} V = 600 \text{ kmol/h} \\ L = 900 \text{ kmol/h} \end{cases}$$

$$y = -\frac{L}{V}x + \frac{F}{V}z$$

$$y = -\frac{900}{600}x + \frac{1500}{600}(0.6)$$

$$y = -\frac{3}{2}x + \frac{3}{2}$$

Same as part A: Same x & y fractions

$$V = 600.0 \text{ kmol/h}$$

$$L = 900.0 \text{ kmol/h}$$

$$x_M = 0.48, x_W = 0.52$$

$$y_M = 0.77, y_W = 0.23$$

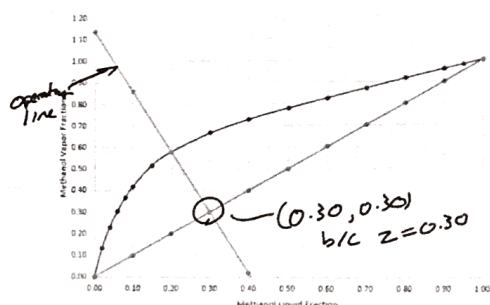
C) If the feed is 30.0% mol methanol & we desire a liquid product that is 20% mol methanol, what $\frac{V}{F}$ must be used? For feed ratio of $100.0 \frac{\text{kmol}}{\text{h}}$, find product flow rates & compositions.

Since feed is 30.0% mol methanol $z = 0.30$, $F = 100.0 \frac{\text{kmol}}{\text{h}}$, and we want $x_M = 0.20$ $\therefore$ according to the data y_M must = 0.579

Step 1 Draw equilibrium diagram w/ data curve & $y=x$ line.

Step 2 We are given z so we know the operating line must intersect at this point. The second intersection point is (x_M, y_M) we know this point b/c $x_M = 0.20$ is given $\therefore$ data tells us that $y_M = 0.579$. Since we have two points we can solve for the slope & intercept of the operating line.

Step 3 Use this line to solve for the necessary variables



$$\text{slope: } \frac{0.579 - 0.30}{0.20 - 0.30} = -2.79 \Rightarrow y = mx + b$$

$$y = -2.79x + b \Rightarrow \text{use } (0.30, 0.30) \Rightarrow 0.30 = -2.79(0.30) + b$$

$$\Rightarrow b = 1.137$$

$$y = -2.79x + 1.137, \text{ since } y = -\frac{L}{V}x + \frac{F}{V}z$$

$$-\frac{L}{V} = -2.79 \Rightarrow L = 2.79V \text{ \& } \frac{F}{V}z = 1.137$$

$$\frac{F(0.30)}{V} = 1.137$$

$$\frac{V}{F} = (3.79)^{-1}$$

